Data Prep for Success with H2O.ai

- Chapter 1: Principles for Machine Learning Data Preparation

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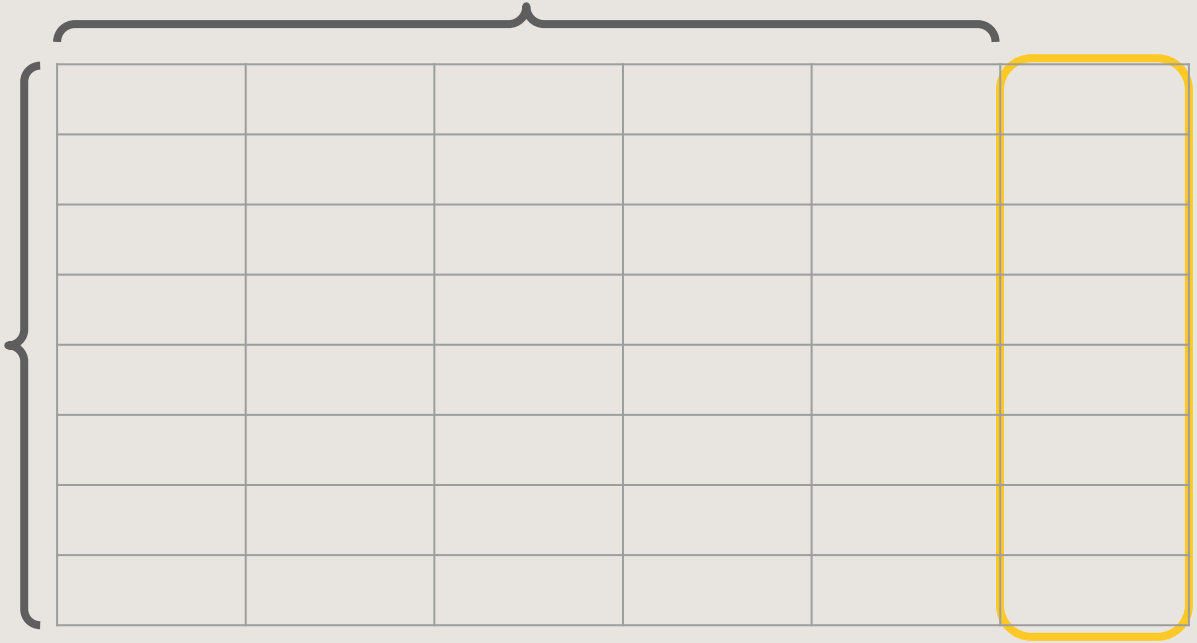
Intro

- *Why is data prep important?*
- *Data Quality Matters -> Gold In - Gold out*
- *The Data Science Lifecycle*

The Tabular Structured Data Format

Columns = Explanatory Variables + Target Variable

Rows = Historical data collected with a pre-defined unit of analysis



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Recap of Modeling Approaches

The **Target Variable** is only needed when we are approaching a Supervised Learning Problem

Supervised Learning: The classical method when we answer the question "What we want to predict?" and we have the historical information for this question collected in the past

Unsupervised Learning: We want to learn patterns from the data without a specific "prediction oriented" question

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Target Definition - Classification

If we want to predict a **condition** the model type is a binary **classification model** with the target variable representing the condition we want to predict with: [True/False, 0/1, Yes/No]

e.g: to build a model to predict fraudulent transactions we have to collect the historical data of transactions and the labeled confirmed frauds

Target Variable

Transaction	Customer	Age	Gender	...	**Is Fraud?**
					No
					No
					Yes

Target Definition - Regression

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If we want to predict a **numerical/quantitative value** the model type is a **regression model**

e.g: to build a model to predict a price to be applied to a product according to its conditions

Target Variable

Product	Brand	Category	Spec XPTO	...	**Price**
					$94.99
					$109.90
					$34

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Unit of Analysis

The Unit of Analysis is the representation of a row in the training dataset

In the two previous examples we have a **Transaction** and a **Product** as our unit of analysis, so for each transaction and product we have a row with their respective additional information

Transaction	Customer	Age	Gender	...	**Is Fraud?**
1					**No**
2					**No**
3					**Yes**

Product	Brand	Category	Spec XPTO	...	**Price**
A					**$94.99**
B					**$109.90**
C					**$34**

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Driverless Example

How the dataset looks like **before** ingesting into Driverless AI

How the dataset looks like **when you are** in Driverless AI

How the dataset looks like after you apply **Custom Recipes** in Driverless AI

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Types of Data Quality Issues that might appear on a dataset

- ***Missing Data***
- *Incomplete Data*
- ***Duplicate Data***
- *Outdated Data*
- ***Data Format Errors***
- *Inaccurate Data*
- *Data Relevancy*
- ***Outliers***

And how to treat some of them into Driverless AI

Enter modification function code

```python
import pandas as pd
X = X.to_pandas()
# Removing NA's for a specific column
X = X.dropna(subset=['ANNUAL_INC'])
# Transforming a specific column to numeric
X['TERM'] = pd.to_numeric(X['TERM'], errors='coerce')
# Sorting the dataset for two specific colums
X.sort_values(by=['LOAN_AMNT', 'ANNUAL_INC'], inplace=True, ascending=False)
# Creating a new rule for a weight column and storing it into a new variable
X['WEIGHT'] = ((X['LOAN_AMNT'] >= 20000)).astype(int)
# Dropping a column
X.drop(columns=['MEMBER_ID'], inplace=True)
# Slicing the data for a specific column
X = X[X['HOME_OWNERSHIP'] != 'OTHER']
return X
```

The two steps (and datasets) for Machine Learning

	Size	Frequency
Step 1 Training	In this step we have a **bigger dataset** (generally thousands or millions of rows), the goal is to **create the model** so we want to inform as much as information as we can	The frequency we create a model is **lower** than we consume Once we have a model trained we can consume with the same model for months or even years
Step 2 Scoring	In this step we have a **smaller dataset** (even a single row), the goal is to **consume the model** so we just want to calculate the prediction(s)	The frequency we consume a model is **greater** than we create It is not unusual to consume a model in a real-time scenario

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The two steps (and datasets) for Machine Learning

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Step 1 Training

Column_A	Column_B	Column_C	Column_D	Column_E	Target

The dataset used for training a supervised model must contain the target variable

Step 2 Scoring

Column_A	Column_B	Column_C	Column_D	Column_E

The dataset used for scoring must contain exactly the **same columns and data types** as the training dataset, except the target

Prediction

The result in this step is exactly to add the prediction column to the initial scoring dataset

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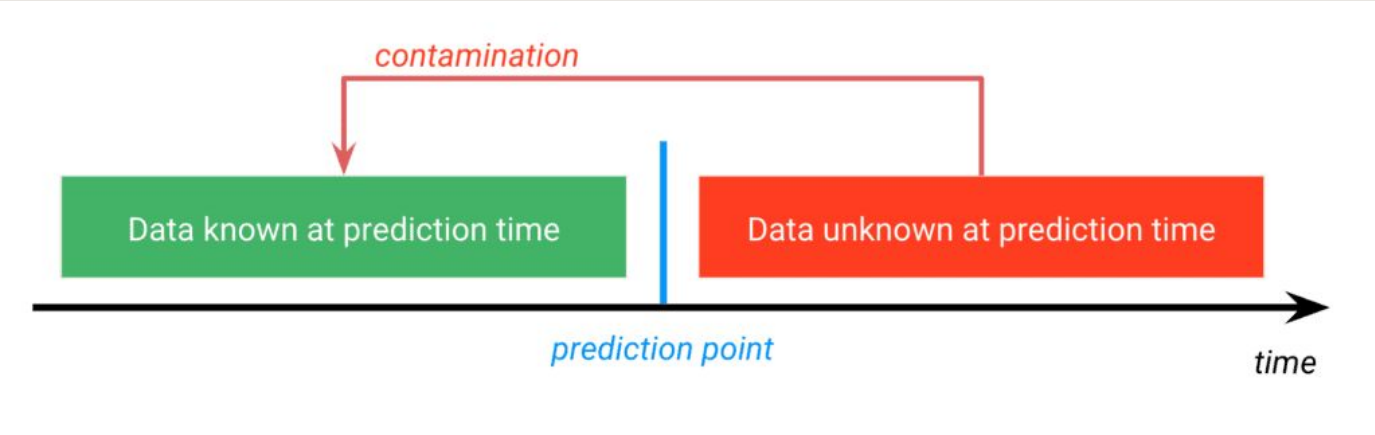
How to Avoid Target Leakage



Data Prep for Success H2O Driverless AI

- Chapter 2: Principles for Time Series Data Preparation

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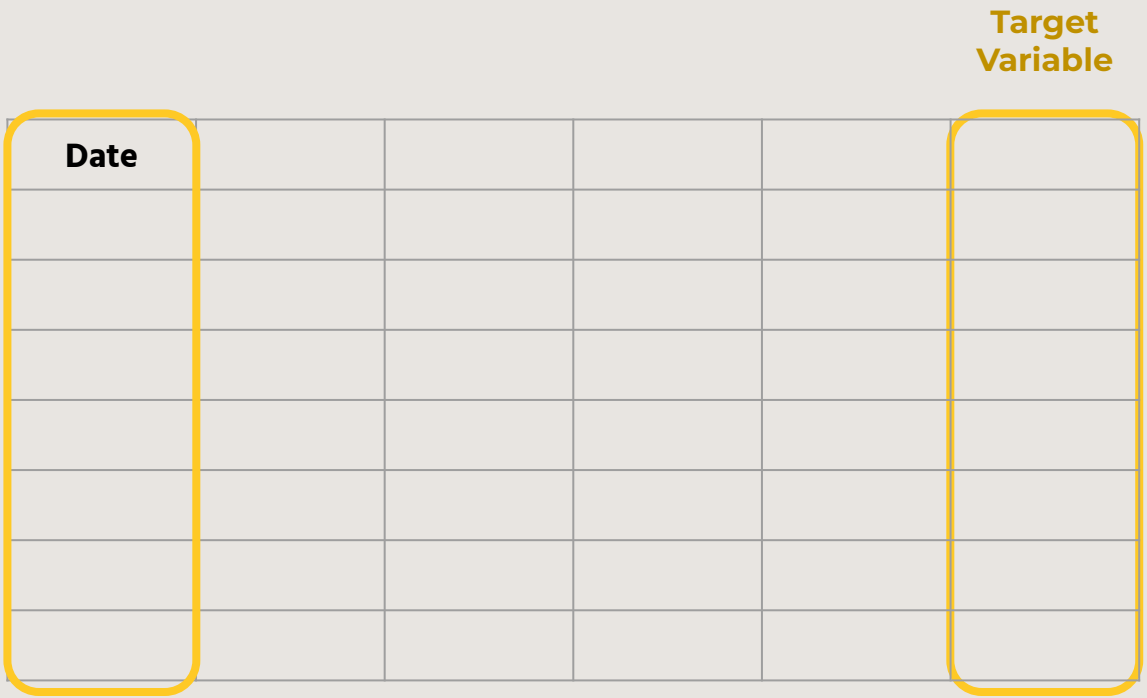
The Tabular Structured Data Format

For Time Series problems the training dataset **MUST** contain a date column with the representation of a timestamp reference for where the target variable happened



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The Tabular Structured Data Format

And because time series are autoregressive problems the **minimal training dataset** can be only the date column plus the target information

Date	Target Variable

Grouped Series

- It is pretty common and it is feasible to handle **multiple series** at the same time and with the same dataset in a Driverless AI experiment
- Multiple series are representations of different series in the same dataset, which means that a specific date can be seen multiple times
- Groups are generally levels or business rules/assumptions. Can be different regions, different product segmentations (brand, category, SKU), different equipments, etc..

Target Variable

Date	Store	Region	
01-01-2023	XYZ	US-1	
01-02-2023	XYZ	US-1	
01-03-2023	XYZ	US-1	
...			
01-01-2023	QWE	US-2	
01-02-2023	QWE	US-2	
01-03-2023	QWE	US-2	

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Best Practices - Grouped Series

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- The better you know the data the better you will have a model performance.
- It is usual when working with multiple series to have hundreds or thousands of series in a single dataset
- But a single experiment to fit every single serie in a single model generally does not work very well
- In this case is recommended to split the dataset into similar groups
  *Business assumptions are always a good call but clustering is a good approach to help define similarity

Target Variable

Date	Store	Region	
01-01-2023	XYZ	US-1	
01-02-2023	XYZ	US-1	
01-03-2023	XYZ	US-1	
...			
01-01-2023	QWE	US-2	
01-02-2023	QWE	US-2	
01-03-2023	QWE	US-2	

Target Variable

Date	Store	Region	
01-01-2023	XYZ	US-1	
01-02-2023	XYZ	US-1	
01-03-2023	XYZ	US-1	

Target Variable

Date	Store	Region	
01-01-2023	QWE	US-2	
01-02-2023	QWE	US-2	
01-03-2023	QWE	US-2	

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Best Practices - Add more information

Date	Group ID	**Number of Events**	**Is Weekend?**	**Target Variable**

Add Calendar Specifics Information

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Best Practices - Add more information

Date	Group ID	**Was sales larger last week?**	**% of ad spending in sales**	**Target Variable**

Add Business Rules Information

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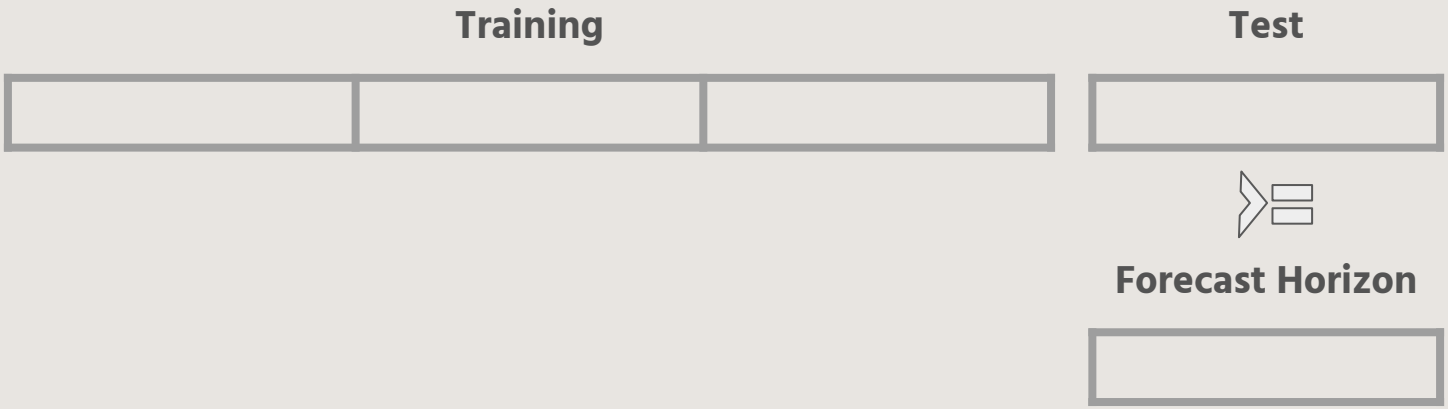
Best Practices - Dataset Size

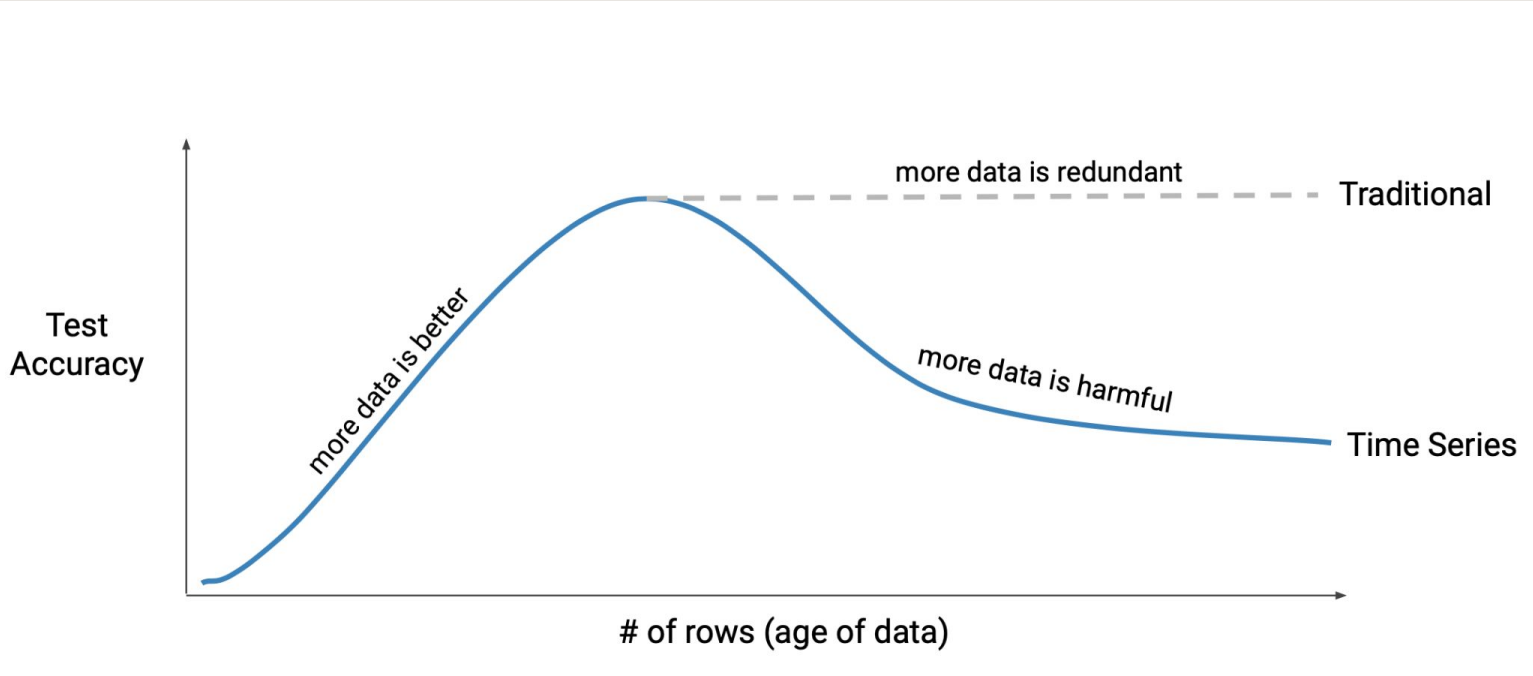
Time Series Constraints

Dataset Size

Usually, the forecast horizon (prediction length) H equals the number of time periods in the testing data N_{TEST} (i.e. $N_{TEST} = H$). You want to have enough training data time periods N_{TRAIN} to score well on the testing dataset. At a minimum, the training dataset should contain at least three times as many time periods as the testing dataset (i.e. $N_{TRAIN} >= 3 \times N_{TEST}$). This allows for the training dataset to be split into a validation set with the same amount of time periods as the testing dataset while maintaining enough historical data for feature engineering.



Best Practices



Best Practices - Reframe the problem

Granularity

- Hourly
- Daily
- Weekly
- Monthly
- Quarterly

- **What is the business question?**
- **Should we predict the mean or the total?**

REMEMBER

MORE granularity

- Less stable target
- More data to train models
- Higher likelihood of capturing interesting dynamics

LESS granularity

- More stable target
- Less data to train models
- Dynamics are often damped
- Tend to be worse ML problems

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Driverless Example

How the dataset looks like before ingesting into Driverless AI

How the dataset looks like when you are in Driverless AI